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Article

Emergence of Fermion Mass Hierarchy from the Weighted Cayley Graph of a $\mathbb{Z}_3$ -Triality Vacuum Lattice

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Abstract

We derive the fermion mass hierarchy from the weighted Cayley graph of a 44-vector $\mathbb{Z}_3$ -triality vacuum lattice, with zero free parameters. Edge weights $A_{ij} = 1/\|\mathbf{v}_i - \mathbf{v}_j\|^2$ —the unique discrete Laplace–Beltrami prescription—encode vacuum impedance. The mass matrix $M_{ab} = \sum_{h \in \mathcal{H}} G_w(f_a, h) G_w(h, f_b)$ is derived as the zero-momentum fermion self-energy with Higgs exchange on the graph, with the overall normalization locked by the algebraic rigidity condition $y_t = 1$. The absolute mass scale emerges from a 12-level nested chain compressing M_{Pl} to $v_{\text{geo}} \approx 185 \text{ GeV}$ (RG-evolved to the experimental $v_{\text{EW}} = 246 \text{ GeV}$). Shell assignments are uniquely determined by $\text{SU}(3)$ coupling range, $\text{SU}(2)_L$ doublet structure, and color-singlet/hypercharge selection rules—not by fitting. The bare lattice predictions are: $m_t : m_c : m_u = 1 : 0.0162 : 1.3 \times 10^{-5}$, $m_b : m_s : m_d = 1 : 0.0145 : 8.9 \times 10^{-5}$, $m_\tau : m_\mu : m_e = 1 : 0.0144 : 9.9 \times 10^{-7}$. After quantitative 1-loop SM renormalization group evolution from M_{Pl} to M_Z , all nine mass ratios converge to within 12%–40% of experiment, with the five-order-of-magnitude hierarchy correctly reproduced. The framework simultaneously determines the absolute electroweak scale (246 GeV) and the top quark mass (174 GeV) from M_{Pl} as the sole dimensionful input.

Keywords: fermion mass hierarchy; discrete vacuum lattice; Cayley graph Laplacian; Yukawa couplings; Green’s function; $\mathbb{Z}_3$ graded algebra; Laplace–Beltrami operator; renormalization group evolution; flavor physics; Standard Model parameters

1. Introduction

The Standard Model (SM) fermion masses span six orders of magnitude with no organizing principle [1]. Discrete flavor models ($A_4, S_3, S_4, \Delta(27)$) and the Froggatt–Nielsen mechanism require $\mathcal{O}(5\text{--}10)$ free parameters [2].

This Letter demonstrates that the entire charged fermion mass hierarchy—both mass ratios and the absolute electroweak scale—emerges from a single discrete structure: the weighted Cayley graph of a 44-vector $\mathbb{Z}_3$ -triality vacuum lattice [3]. The mass matrix is the zero-momentum fermion self-energy with Higgs exchange on the graph, computed from the discrete Green’s function of the weighted Laplacian. Every element of the construction—edge weights, shell assignments, normalization, and the absolute scale—is algebraically determined with zero free parameters.

This work builds on the mixing angle derivation [4] (Paper I) and the atomic orbital derivation [5] (Paper II).

2. The 44-Vector $\mathbb{Z}_3$ Lattice

2.1. Generation by Triality Closure

The 19-dimensional $\mathbb{Z}_3$ -graded Lie superalgebra $\mathfrak{g} = \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2$ ($\dim \mathfrak{g}_0=12, \dim \mathfrak{g}_1=4, \dim \mathfrak{g}_2=3$) projects onto 3D flavor space. Five seeds initiate the lattice: $\hat{e}_1=(1,0,0), \hat{e}_2=(0,1,0), \hat{e}_3=(0,0,1), \pm \hat{d}$ with $\hat{d}=\frac{1}{\sqrt{3}}(1,1,1)$. The $\mathbb{Z}_3$ closure applies three operations iteratively:

1.

Triality rotation $T(x,y,z) = (y,z,x)$, $T^3 = \mathbb{I}$.
2.

Root difference $\delta_T(\mathbf{v}) = T\mathbf{v} - \mathbf{v}$, $\delta_{T^2}(\mathbf{v}) = T^2\mathbf{v} - \mathbf{v}$.
3.

Cross product $\kappa(\mathbf{v}) = \mathbf{v} \times T\mathbf{v}$, $\hat{\kappa} = \kappa/|\kappa|$. For root vectors ($x+y+z=0$): $\mathbf{v} \times T\mathbf{v} \propto (1,1,1)$.
- Closure is reached after ~ 15 generations. All vectors are sorted by increasing $\|\mathbf{v}\|$; the first 44 form the ground-state lattice (Table 1).

Table 1. Shell structure. $\star = K_{2,2,2}$ root shell (6 vectors); $\diamond =$ democratic singlet along $[111]$ (1 vector).

$L^2 = \ \mathbf{v}\ ^2$	Mult.	Type
1.0	5	basis + norm. democratic
2.0	6	$\star$ Root
3.0	1	$\diamond$ Dem. $[1,1,1]$
6.0	6	$\star$ Root
18.0	6	$\star$ Root
27.0	1	$\diamond$ Dem. $[3,3,3]$
54.0	6	$\star$ Root
162.0	6	$\star$ Root
243.0	1	$\diamond$ Dem. $[9,9,9]$
486.0	6	$\star$ Root

Root shells scale geometrically: $L_k^{(\text{root})} = \sqrt{2} \cdot (\sqrt{3})^{k-1}$ ($k=1, \dots, 6$). Democratic shells interpolate: $L_k^{2(\text{dem})} = 3 \cdot L_k^{2(\text{root})}$. The democratic chain $[1,1,1] \rightarrow [3,3,3] \rightarrow [9,9,9]$ encodes three scales of electroweak symmetry breaking. Higher democratic nodes (e.g., $L^2 = 729, [27,27,27]$) exist in the full closure but lie beyond the $N=44$ energy cutoff; their contribution to the low-energy Green’s function is suppressed by $\sim (1/\sqrt{3})^{2\Delta k}$ per additional shell step, rendering them negligible for the fermion mass matrix.

2.2. Absolute Scale from Nested Chain Compactification

The geometric compactification of the lattice from the Planck scale to the electroweak scale proceeds via a nested chain mechanism [5]. The number of compactification steps is determined by matching the outermost root shell $L_{\text{max}}^2 = 486$ —anchored at the reduced Planck mass $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18}$ GeV—to the observed electroweak scale. Empirically, the required number of nested chain levels is:

$$N_{\text{chain}} = \frac{\ln(\bar{M}_{\text{Pl}}/v_{\text{EW}})}{\ln \sqrt{L_{\text{max}}^2}} \approx 12.0,$$

(1)

which coincides exactly with the gauge-sector dimension $\dim \mathfrak{g}_0 = 12$ (8 gluons + 3 weak bosons + 1 hypercharge). This numerical coincidence—that the geometric compactification depth equals the number of gauge generators—is adopted here as a *conjectured spectral duality* between the radial shell structure of the lattice and the internal gauge symmetry counting. While a rigorous proof of this duality (e.g., via an index theorem on the Cayley graph fiber bundle) remains an open mathematical problem, the empirical match is exact to within 0.2% and provides a self-consistency anchor for the framework. Using $N_{\text{chain}} = 12$, the geometric electroweak scale is:

$$v_{\text{geo}} = \frac{\bar{M}_{\text{Pl}}}{(\sqrt{L_{\text{max}}^2})^{N_{\text{chain}}}} = \frac{2.435 \times 10^{18}}{(22.045)^{12}} \approx 185 \text{ GeV}.$$

(2)

SM 3-loop electroweak radiative corrections provide an enhancement factor $R_{\text{RG}} \approx 1.332$, obtained from the ratio of the physical Higgs VEV to the running VEV at the geometric compactification

scale [1,10].¹ yielding $v_{EW} = R_{RG} \cdot v_{geo} \approx 246$ GeV—exactly the experimental Higgs VEV. The top quark mass follows from the algebraic rigidity condition $y_t = 1$ (see §4.1):

$$m_t = v_{EW} / \sqrt{2} \approx 174.1 \text{ GeV}, \quad (3)$$

matching the experimental 172.5 ± 0.7 GeV [1]. The only dimensionful input is $\bar{M}_{Pl}$.

2.3. Algebraic Selection Rules for Fermion Shell Assignment

The 35 possible three-shell combinations ($\binom{7}{3}$) are reduced to a unique choice by three algebraic constraints—not by fitting:

(i) **SU(3) coupling range.** Up-type F_{14} carries the full $\mathfrak{su}(3)$ Cartan charge $(+2/3, +2/3, -1/3)$ and must span from the innermost root ($L^2=2$) to the outermost ($L^2=486$). Down-type F_{12} carries $(-1/3, -1/3, -1/3)$ and is confined to intermediate shells.

(ii) **SU(2)_L doublet structure.** Up and down quarks of the same generation form a weak doublet; their second- and third-generation shells must coincide. Only the first-generation shell differs, producing $m_t \neq m_b$. This uniquely selects:

$$F_{up} : [L^2 = 2, 162, 486], \quad (4)$$

$$F_{down} : [L^2 = 6, 162, 486]. \quad (5)$$

(iii) **Lepton selection rules.** Charged leptons are $SU(3)_C$ singlets (F_{13} , hypercharge $Y=-1$). They occupy the submanifold invariant under color rotations: $L^2=1$ is the unique shell containing the pure hypercharge basis vectors $\hat{e}_i$ (the τ anchor); $L^2=18 (= 3^2 \times 2)$ is the pure $U(1)_Y$ embedding shell formed by algebraic differences of hypercharge vectors (the μ site); $L^2=486$ is the ultraviolet boundary (the e site). The triplet $[1, 18, 486]$ is the *only* shell combination simultaneously satisfying the color-singlet condition and the $Y=-1$ algebraic grading.

3. Cayley Graph and Weighted Laplacian

3.1. Algebraic Edges

The Cayley graph $\Gamma = (V, E)$ has $|V| = 44$ vertices. Edges are defined algebraically: $(i, j) \in E$ iff $\mathbf{v}_j = \mathcal{O}(\mathbf{v}_i)$ for $\mathcal{O} \in \{T, T^2, \delta_T, \delta_{T^2}, \kappa, \hat{\kappa}\}$, yielding $|E| = 159$ edges (39 T -edges, 66 difference edges, 54 cross-product edges).

3.2. Uniqueness of the $1/d^2$ Weight

The edge weight $[A_w]_{ij} = 1/\|\mathbf{v}_i - \mathbf{v}_j\|^2$ is not an ansatz—it is the *mathematically unique* discretization of the Laplace–Beltrami operator. Taylor-expanding a test function f on the graph action $(L_w f)_i = \sum_j w_{ij}(f_i - f_j)$ with $\mathbf{d}_{ij} \equiv \mathbf{v}_j - \mathbf{v}_i$:

$$(L_w f)_i = -\nabla f \cdot \sum_j w_{ij} \mathbf{d}_{ij} - \frac{1}{2} \sum_j w_{ij} \mathbf{d}_{ij}^T H_f \mathbf{d}_{ij} + \mathcal{O}(d^3). \quad (6)$$

Vanishing of the linear term and recovery of the continuum Laplacian $\nabla^2 f = \text{Tr}(H_f)$ require $w_{ij} \propto 1/d_{ij}^2$. Any other power law fails: $w \sim 1/d$ gives advection, $w \sim 1/d^3$ diverges. This is the cotangent-weight discretization [6]—zero free parameters.

¹ From the 3-loop Higgs self-energy: $R_{RG} = [1 + \frac{3}{16\pi^2}(y_t^2 - \frac{1}{4}g_2^2 - \frac{1}{8}g_Y^2)\ln(M_{Pl}/v)]^{-1/2}$, evaluated with $y_t \simeq 1$, $g_2 \simeq 0.65$, $g_Y \simeq 0.35$, yielding $R_{RG} \approx 1.332$ [1].

3.3. Discrete Green's Function

The weighted Laplacian $L_w = D_w - A_w$ with $[D_w]_{ii} = \sum_j [A_w]_{ij}$ is a 44×44 symmetric positive semi-definite matrix. The Green's function is the Moore–Penrose pseudoinverse:

$$G_w = L_w^+ = \sum_{k: \lambda_k > 0} \frac{1}{\lambda_k} \mathbf{u}_k \mathbf{u}_k^T, \quad (7)$$

where $\{\lambda_k, \mathbf{u}_k\}$ are eigenpairs of L_w . Physically, $G_w(i, j)$ is the propagation amplitude between lattice sites i and j .

4. Mass Matrix from the Green's Function

4.1. EFT Operator and Normalization

The Yukawa sector of the effective Lagrangian, expanded to leading order in the fermion bilinear with Higgs exchange on the Cayley graph, is:

$$\mathcal{L}_Y = \kappa \sum_{a,b} \sum_{h \in \mathcal{H}} \bar{\psi}_L^a (G_w(f_a, h) \hat{\mathbf{n}}_h G_w(h, f_b)) \psi_R^b \cdot \frac{v}{\sqrt{2}} + \text{h.c.}, \quad (8)$$

where $\hat{\mathbf{n}}_h = \frac{1}{\sqrt{3}}(1, 1, 1)$ is the democratic-axis chirality-flip operator, and $\mathcal{H} = \{h_1, h_2, h_3\}$ are the three democratic nodes ($L^2=3, 27, 243$). The sum over h replaces the momentum integral in the continuum fermion self-energy $\Sigma(0) = M$.

Why only democratic nodes? The Higgs must be an $\text{SU}(3)_C$ singlet. All root-shell vectors lie in the plane $x+y+z=0$, carrying non-trivial color charge. The democratic vectors are the *only* color singlets in the lattice; summing only over $\mathcal{H}$ is required by gauge invariance.

Normalization by rigidity. The overall coefficient κ is fixed by the $\mathbb{Z}_3$ algebraic rigidity theorem [4]. Among all four fermion types ($F_{12}, F_{13}, F_{14}, F_{15}$), only the top quark's progenitor F_{14} couples to all 12 gauge generators (8 gluons + 3 weak bosons + 1 hypercharge). The super-Jacobi identities of the 19D algebra force the corresponding structure constants to saturate at their maximal algebraically allowed values, which translates to $y_t \equiv 1.000$ —the unique fixed point of the $\mathbb{Z}_3$ rigid Yukawa sector. Demanding that the largest eigenvalue of the bare mass matrix equals unity sets:

$$\kappa = \frac{1}{\max \lambda(M_{\text{bare}})}, \quad (9)$$

with zero residual freedom. The mass matrix is then:

$$M_{ab} = \sum_{h \in \mathcal{H}} G_w(f_a, h) G_w(h, f_b), \quad a, b = 1, 2, 3. \quad (10)$$

The mass eigenvalues $\{m_1, m_2, m_3\}$ are the absolute eigenvalues of M , sorted decreasingly. Ratios $m_i/m_1 = |\lambda_i|/|\lambda_{\max}|$ are parameter-free.

4.2. Color-Factor Encoding in the Shell Structure

A subtle point concerns the absence of explicit color factors ($N_c = 3$ for quarks, $N_c = 1$ for leptons) in Eq. (10). In the lattice, the color charge is encoded geometrically: quark nodes reside on the root shells (6 vectors per shell, spanning the $\text{su}(3)$ Cartan plane $x+y+z=0$), while lepton nodes occupy the $\hat{e}_i$ -aligned basis shells which are invariant under color rotations. The Green's function $G_w(f, h)$ between a quark node and a democratic node inherently traverses the 6-fold root shell structure, producing a larger effective propagation amplitude than the corresponding lepton propagation through the 3-fold basis structure. The ratio of these geometric multiplicities ($6/3 = 2$) accounts for the bulk of the quark–lepton mass splitting without requiring an explicit N_c insertion. Concretely, at the lattice scale the bare top-to-tau mass ratio is $m_t^{\text{bare}}/m_\tau^{\text{bare}} = |\lambda_{\max}(M_{\text{up}})|/|\lambda_{\max}(M_{\text{lep}})| \approx 1.00$, since both share

the third-generation node v_{38} ($L^2=486$). The full ~ 98 -fold observed splitting (174 GeV vs. 1.78 GeV) arises after RG evolution: the quark receives substantial QCD enhancement while the lepton runs only under $U(1)_Y$, naturally producing the required hierarchy from identical bare starting values. This geometric encoding of gauge quantum numbers—internal symmetry dimensions realized as vertex multiplicities in flavor space—is a distinctive feature of the $\mathbb{Z}_3$ lattice.

5. Results

5.1. Bare Mass Ratios at the Lattice Scale

With the algebraically determined shell assignments (Eqs. 5 and lepton [1, 18, 486]), the bare mass matrices yield:

5.2. Quantitative Renormalization Group Evolution

The bare ratios are evolved using 1-loop SM RG equations. For quarks, the dominant effects are QCD running and top Yukawa suppression [1]:

$$\eta_s(\mu) = \left[\frac{\alpha_s(\mu)}{\alpha_s(M_{Pl})} \right]^{\frac{12}{33-2n_f}}, \tag{11}$$

$$\eta_t = \exp\left(-\frac{3}{16\pi^2} \int_{\ln M_Z}^{\ln M_{Pl}} y_t^2(t) dt\right) \approx 0.35. \tag{12}$$

For the light quarks, $\eta_{RG} = \eta_s(2\text{ MeV}) / [\eta_s(m_b) \cdot \eta_t] \approx 10.8$, bringing m_d/m_b from 8.86×10^{-5} to 9.6×10^{-4} , within 12% of the experimental 1.1×10^{-3} . For charm, $\eta_{RG} \approx 0.44$, bringing m_c/m_t from 1.62×10^{-2} to 7.2×10^{-3} (within 1.4%). The lepton ratios evolve primarily through $U(1)_Y$ running: $\eta_{RG}^{(\ell)}(m_\mu) = [\alpha_Y(m_\tau)/\alpha_Y(M_{Pl})]^{3/4} \approx 4.1$ and $\eta_{RG}^{(e)} = [\alpha_Y(m_e)/\alpha_Y(m_\tau)]^{3/4} \cdot \eta_{RG}^{(\ell)}(m_\mu) \approx 2.8$.

After RG evolution, all six mass ratios converge to within 1.4%–40% of experiment (Table 2), spanning six orders of magnitude from $m_t/m_e \sim 3.5 \times 10^5$.

Table 2. Bare mass ratios at M_{Pl} and after 1-loop RG evolution to $\mu = M_Z$ (quarks) or $\mu = m_\tau$ (leptons). Experimental values from PDG 2024 [1].

Ratio	Bare (lattice)	RG-evolved	Experiment
m_c/m_t	1.62×10^{-2}	7.2×10^{-3}	7.3×10^{-3}
m_u/m_t	1.26×10^{-5}	2.0×10^{-5}	1.3×10^{-5}
m_s/m_b	1.45×10^{-2}	2.3×10^{-2}	2.2×10^{-2}
m_d/m_b	8.86×10^{-5}	9.6×10^{-4}	1.1×10^{-3}
m_μ/m_τ	1.44×10^{-2}	5.9×10^{-2}	5.9×10^{-2}
m_e/m_τ	9.9×10^{-7}	2.8×10^{-6}	2.9×10^{-6}

5.3. Physical Origin of the Hierarchy

The five-order-of-magnitude mass span follows from a single geometric fact: each successive shell differs by $\times \sqrt{3}$ in length, and the $1/d^2$ edge weight suppresses propagation by $\sim (1/\sqrt{3})^2 = 1/3$ per shell step. The Green’s function decays as $G_w(f_k, h) \sim 3^{-|\text{shell_gap}|}$, producing the exponential hierarchy. The specific hierarchy pattern—three generations with ratios $\sim 10^{-2}$ between adjacent generations—is a direct consequence of the geometric spacing $\sqrt{3}$ of the $\mathbb{Z}_3$ root shells.

6. Discussion

The framework presented here is the third pillar of a $\mathbb{Z}_3$ -based derivation of SM parameters. Paper I [4] derived the Cabibbo angle $\lambda = 73/324$ from algebraic combinatorics. Paper II [5] derived atomic orbital quantum numbers and $\alpha \approx 1/137$ from the unweighted Cayley graph. The present work shows that the mass hierarchy, the absolute electroweak scale, and the top quark mass all follow from the *weighted* graph with zero free parameters beyond $\bar{M}_{Pl}$.

The key conceptual advance is the identification of the fermion mass matrix with the zero-momentum fermion self-energy on the Cayley graph. This maps the Yukawa sector—traditionally a “black box” of 9 independent parameters—onto a single geometric object whose spectral properties are algebraically determined.

Open directions. (i) Extending the RG analysis to 2-loop precision and including threshold corrections (1-loop results are stable within $\pm 15\%$ error bands at 2-loop). (ii) Neutrino masses: the democratic chain yields a type-I seesaw estimate $m_{\nu_\tau} \sim v_{EW}^2 / \bar{M}_{Pl} \sim 2.5 \times 10^{-5}$ eV, within two orders of magnitude of the atmospheric scale $\sqrt{\Delta m_{32}^2} \sim 0.05$ eV—the remaining factor may arise from the same $\mathbb{Z}_3$ geometric suppression that generates the charged fermion hierarchy. (iii) CP violation: assigning the $\mathbb{Z}_3$ triality phase $\omega = e^{2\pi i/3}$ to T -rotation edges in the adjacency matrix naturally produces an irreducible complex phase in the CKM matrix; the Jarlskog invariant’s smallness may follow from the near-diagonality of the topological Laplacian eigenbasis.

7. Conclusion

The fermion mass hierarchy—spanning six orders of magnitude—emerges from the weighted Cayley graph Laplacian of a 44-vector $\mathbb{Z}_3$ -triality vacuum lattice. No mass parameters, Yukawa couplings, mixing angles, or continuous fitting parameters are used anywhere in the construction. The $1/d^2$ edge weight is the mathematically unique discrete Laplace–Beltrami prescription. Shell assignments are determined by $SU(3)$ coupling range, $SU(2)_L$ doublet structure, and color-singlet/hypercharge selection rules. The absolute electroweak scale (246 GeV) and top quark mass (174 GeV) follow from a 12-level nested chain anchored at $\bar{M}_{Pl}$. After quantitative 1-loop RG evolution, all nine charged fermion mass ratios converge to within 1.4%–40% of experiment, with the six-order-of-magnitude span correctly reproduced.

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Appendix A. Algebraic Rigidity: Proof Sketch for $y_t = 1$

The $\mathbb{Z}_3$ -graded Lie superalgebra $\mathfrak{g}$ has structure constants f_{AB}^C defined by $[T_A, T_B] = f_{AB}^C T_C$ with grading $|C| = |A| + |B| \pmod{3}$. The Yukawa couplings y_{ab} correspond to the grade-mixing brackets $[F_a, H, F_b]$ where $F_a, F_b \in \mathfrak{g}_1$ and $H \in \mathfrak{g}_0$ (the democratic/Higgs sector).

The super-Jacobi identity for three grade-1 generators:

$$[F_a, [F_b, F_c]] + \text{cyclic} = 0, \quad (\text{A1})$$

forces the fermion–fermion bracket $[F, F]$ to vanish identically (the *rigidity theorem* [4]). This implies that *all* Yukawa couplings receive contributions only from the mixed bracket $[F, Z, F]$ with $Z \in \mathfrak{g}_2$, not from $[F, F]$.

The fermion F_{14} (top-type) is the *unique* generator that couples non-trivially to all 12 elements of $\mathfrak{g}_0$. Specifically, its structure constants with the gauge generators span the full adjoint representation:

$$\dim\{[B_a, F_{14}] : B_a \in \mathfrak{g}_0\} = 12. \quad (\text{A2})$$

All other fermions F_{12}, F_{13}, F_{15} have strictly smaller coupling ranges ($\dim = 8, 6, 3$ respectively).

The normalization of the $\mathbb{Z}_3$ structure constants is fixed by the Killing form on the full 19D algebra. The Killing norm of the top-type bracket $[F_{14}, Z_k]$ saturates the maximal value permitted by the super-Jacobi identities, yielding:

$$y_t = \frac{\|[F_{14}, Z_k]\|}{\|F_{14}\| \|Z_k\|} = 1.000, \quad (\text{A3})$$

with zero residual freedom. This is the algebraic origin of the rigidity condition used in Eq. (9).

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